Higher National Diploma in Engineering (Civil)

Third Year, Second Semester Examination - 2020

CE 3117 - Concrete Technology

(01.)

I. What are the main constituents of cement. (04 Marks)

II. Name four properties that an aggregate should have in order to use it as an ingredient of concrete. (04 Marks)

III. Describe the importance of Aggregate crushing test. (04 Marks)

IV. Briefly explain the procedure to determine consistency of cement in the laboratory. (04 Marks)

V. State the difference between Initial setting time and Final setting time in a cement paste. (04 Marks)

(02.)

I. Name four advantages of using Fly Ash in Concrete mixture. (04 Marks)

II. State the drawbacks of using Marine-dredged aggregate as concrete ingredient. (04 Marks)

III. Admixtures are used in concrete for various purposes and that usage is beneficial for several parties involved in construction. Rationalize this statement. (04 Marks)

IV. Write short notes on following admixtures. (08 Marks)

a) Air entraining admixtures

b) Accelerating admixtures

c) Water-reducing admixtures (Plasticizers)

d) Corrosion inhibitors

(03.)

I. List out four methods that can be used for concrete transportation. (04 Marks)

II. Briefly describe two methods of concrete batching? (04 Marks)

III. Briefly explain the term "Concrete Durability". (02 Marks)

IV. Name four factors that affect the durability of concrete. (04 Marks)

V. Concrete compaction may be associated with various defects in concrete structures. Explain this statement in terms of under-vibration and over-vibration. (06 Marks)

(04.)

I. State four effects of corrosion on steel reinforcement. (04 Marks)

II. What are the methods that you can recommend to avoid or delay corrosion of steel reinforcement? (04 Marks)

III. Define the term "Fatigue failure". (03 Marks)

IV. Write down short notes on following concrete deterioration methods.

a) Freezing and thawing

b) Alkali-Silica reaction

c) Delayed ettringite formation

(03 x 03 Marks)

(05.)

I. What is the importance of maintaining workmanship in concrete related construction? (04 Marks)

II. What are the advantages of post-tensioning concrete over the pre-tensioning concrete? (04 Marks)

III. Describe three factors influencing the concrete fire resistance. (06 Marks)

IV. Briefly describe the following concrete curing methods. (06 Marks)

a) Water curing

b) Membrane curing

c) Steam curing

(06.)

I. Name two types of concrete related field tests that you practice at the construction site as a trainee engineer. Mention the importance of the tests you named. (04 Marks)

II. What are the requirements of materials that are used for formwork construction (04 Marks)

III. Briefly describe the purpose of mix design. (04 Marks)

IV. Find mix proportions for the grade 30 concrete (The specific strength) having workability of 5 S from Vee-bee test. Used cement type is O.P.C. (strength class 42.5). Maximum size of aggregate is 20 mm. Coarse aggregates uncrushed and fine aggregates crushed with fine aggregate percentage of passing 600mum sieve is estimated as 50% from the total aggregate content. Relative density for combined aggregate density is 2.65. Assume 2.5% defective and Standard Deviation is 4. Maximum Water/Cement ratio is 0.5 and minimum cement content is 350kg/m^3 as per the specification of BS 8110: Part I.

Use the following tables and graphs for calculations. 83 (08 Marks)

Table 01: Probability factor

Defectives %	**K value**
10	1.28
5	1.64
2.5	**1.96**
1	2.33

Table 02: Compressive strength of concrete mixes made with a free-water cement ratio of 0.5

Cement strength class	*Type of coarse aggregate*	*Compressive strength (N/mm²)*			
		Age (days)			
		3	*7*	*28*	*91*
42.5	Uncrushed	22	30	42	49
	Crushed	27	36	49	56
52.5	Uncrushed	29	37	48	54
	Crushed	34	43	55	61

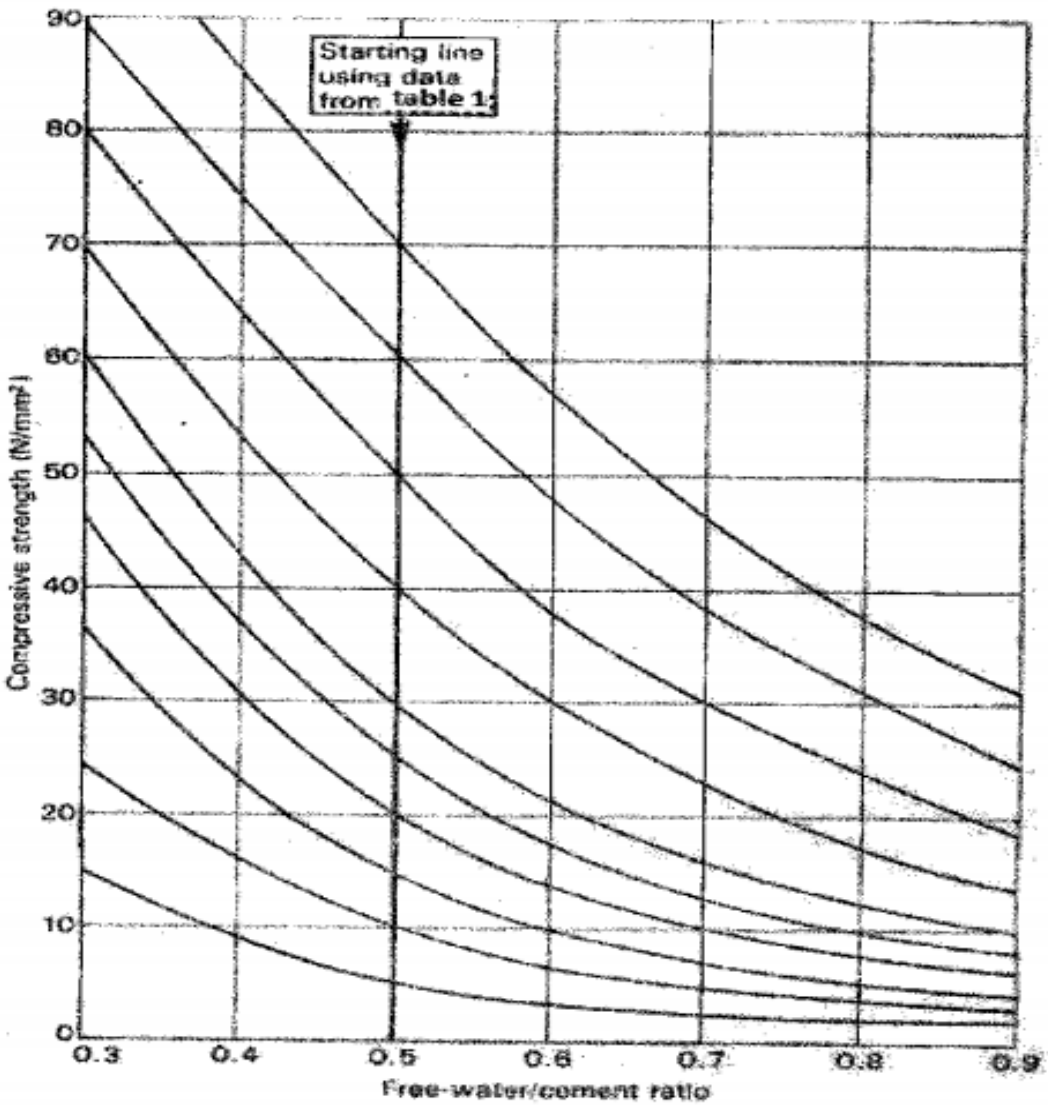

Figure 01: Relation between compressive strength and free w/c ratio

Table 3: Approximate free water content required to give various levels of workability

Maximum size of aggregate (mm) (in)	Slump (mm)	0-10 > 12	10-30	30-60	60-180
	V-B (sec)		6-12	3-6	0-3
	Type of aggregates				
10(3/8)	Uncrushed	150	180	205	225
	Crushed	180	205	230	250
20(3/4)	Uncrushed	135	160	180	195
	Crushed	170	190	210	225
40(5/2)	Uncrushed	115	140	160	175
	Crushed	155	175	190	205

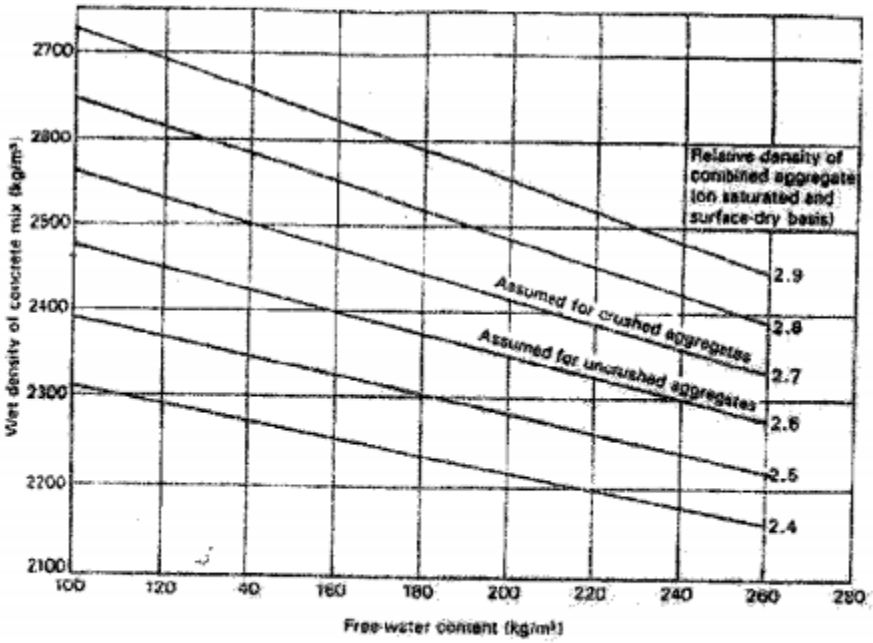

Figure 02: Estimated wet density for fully compacted concrete

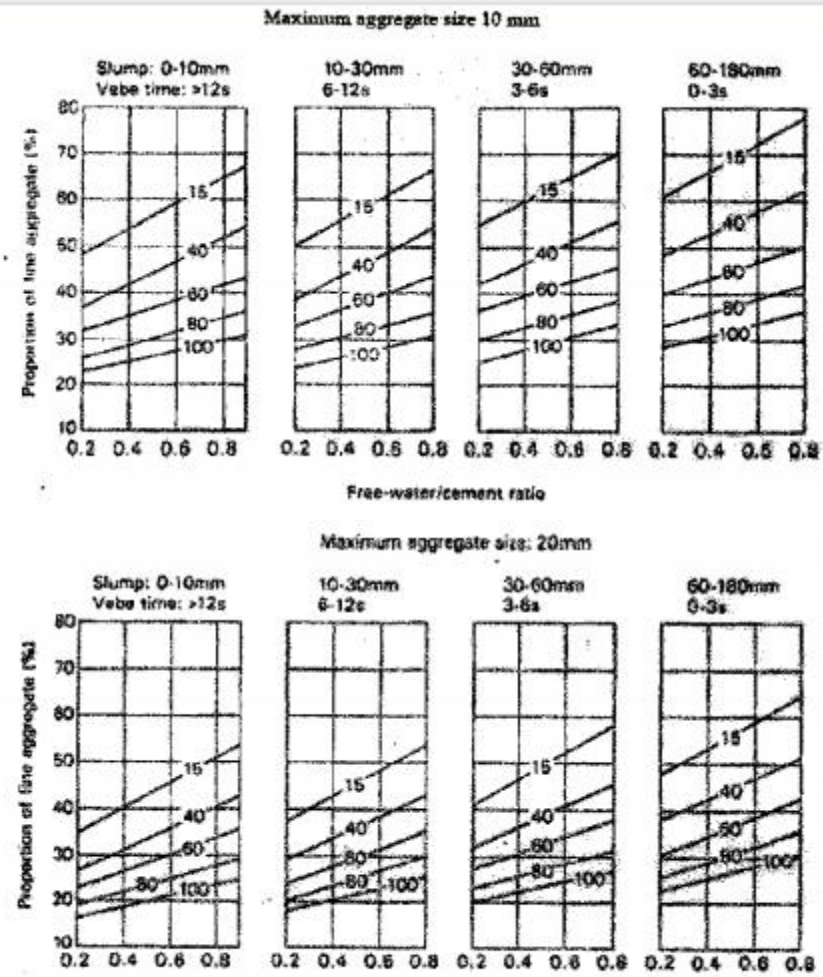

Figure 03: Recommended % of fine aggregate as a function of free w/c ratio for various values of workability and mix